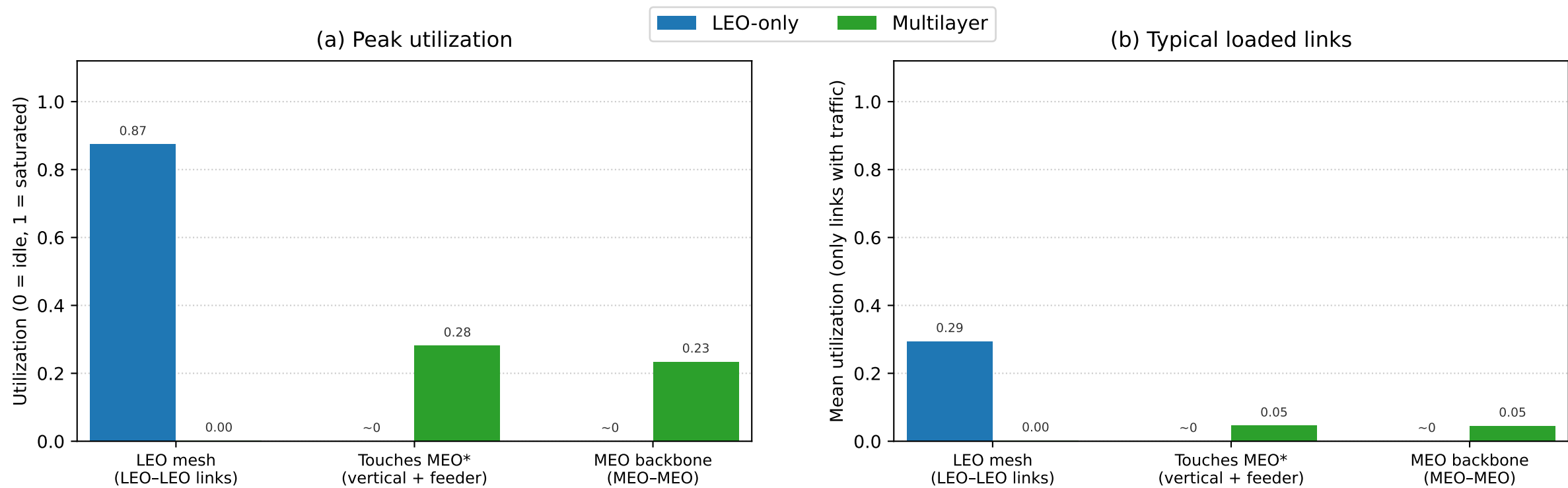


Figure J — Traffic moves off the LEO mesh onto MEO (Kuiper, 3 flows averaged) — 25 s sim, 1000 ms state updates



KEY RESULT
Peak on LEO mesh: LEO-only ≈ 0.87 , multilayer ≈ 0.00 . Relative drop: $\approx 100\%$.
Typical busy mesh link: LEO-only ≈ 0.29 , multilayer ≈ 0.00 . Relative drop: $\approx 99\%$.

WHY THE GREEN LEFT BARS ARE SMALL: Multilayer routing uses MEO, so the heaviest LEO-LEO hop is much less loaded — LEO satellites are still there; congestion on the mesh is what eased.

WHY THE GREEN MIDDLE/RIGHT BARS EXIST: That is traffic on links involving MEO — the intended offload. Good for spreading load, but MEO can become congested if demand grows.

SCOPE: Numbers come from `isl_utilization.csv` (whole network), not from tracing one flow end-to-end.

* “Touches MEO” = any logged ISL with a MEO satellite on one end (includes LEO-MEO, MEO-ground feeders, etc.). “Relative drop” on the mesh group: (LEO-only bar – Multilayer bar) / LEO-only bar, after averaging the three runs.